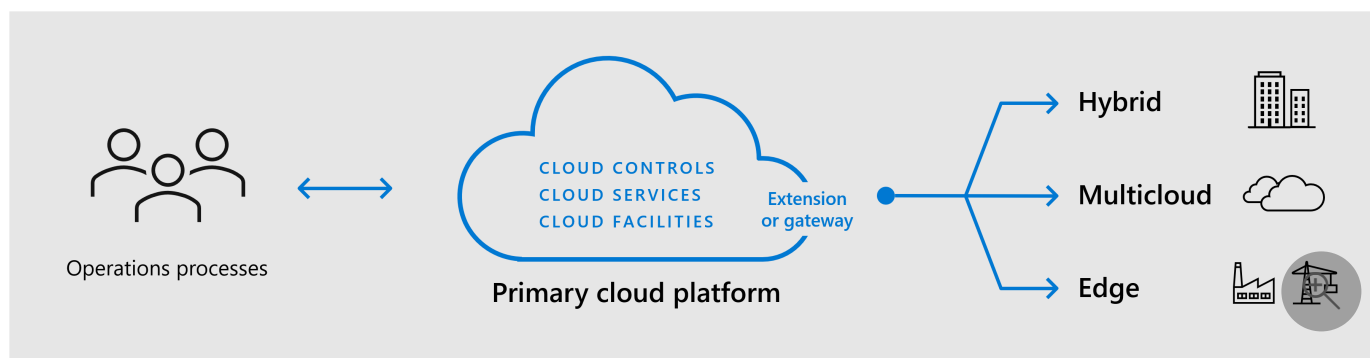


Unified hybrid and multicloud operations

Hybrid cloud refers to a mix of on-premises/private infrastructure and public cloud services working together, while multicloud means using multiple cloud providers concurrently. Many enterprises today have siloed teams, distributed sites, and systems spread across on-premises datacenters and various clouds. The challenge is to unify these environments in a secure, well-managed way that enables modernization from cloud to edge. This guidance provides a prescriptive end-to-end framework for unifying and managing hybrid and multicloud environments with Azure as the central control plane.

Quickstart: [Azure hybrid and multicloud services](#)



We explain how Azure solutions, like Azure Arc, Azure Monitor, Azure Kubernetes Service, Microsoft Fabric, Azure IoT, and Azure Local, help unify control and modernize IT across all environments. The goal is to establish a standard operational model that breaks down silos and delivers consistent practices everywhere. We detail how to secure, manage, and modernize resources from cloud to edge, unifying previously isolated teams and systems under one Azure-driven strategy.

The first step is to establish a clear strategy for hybrid and multicloud that aligns with your business goals and emphasizes unification. This alignment ensures business value (agility, resilience, cost optimization, innovation) drives your cloud adoption rather than ad-hoc decisions. Key activities in this phase include defining drivers and vision, setting guiding principles, determining your cloud mix, and mapping Azure services to objectives.

1. Define business drivers for hybrid and multicloud

Start by identifying why your organization is adopting hybrid and multicloud. Business drivers provide focus and ensure the approach delivers measurable value. These drivers must link to

concrete business outcomes or key performance indicators (KPIs) to guide decisions and prevent fragmented technology choices. Common drivers include:

- **Vendor flexibility:** Reduce dependence on any single provider to mitigate lock-in and allow cost optimization and best-of-breed service use. For example, avoid relying solely on one cloud's features if an alternative could save cost or offer unique capabilities.
- **Business unit diversity:** Accommodate different teams, acquisitions, or subsidiaries that use various platforms. A unified strategy prevents operational silos and establishes central governance across all environments while respecting existing investments.
- **Compliance and data residency:** Meet regulatory mandates for data sovereignty or specific security controls. Use Azure Local, which Azure Stack HCI is now a part of, for workloads that must remain on-premises. Connect to Azure cloud services for other data and applications.
- **Resilience and disaster recovery:** Improve availability by distributing workloads and establishing multi-environment failover. Resilience sustains operation during localized faults; disaster recovery restores normal operations after broader incidents. Design recovery between Azure and secondary environments (other clouds or on-premises) with unified processes to ensure minimal downtime.
- **Performance optimization:** Place workloads closer to users or data sources to reduce latency. For example, deploy Azure Local instances in factory locations for real-time processing, while maintaining centralized coordination through Azure cloud services.
- **Modernization and innovation:** Use specialized cloud services to drive transformation. For example, use Foundry Tools and Microsoft Fabric analytics while integrating data and apps across other clouds as needed.
- **Unifying silos:** Eliminate internal silos between infrastructure, cloud, and application teams. Establish common tools and processes, like Azure Arc for resource management, Azure Monitor for observability, to create shared visibility and collaboration across formerly isolated groups.

For each business driver, link it to a specific business outcome or KPI. For example, if a driver is "avoid downtime," an outcome could be achieving 99.99% availability. If "support global expansion" is a driver, an outcome might be launching in some new geographies within a year. If unifying silos is a goal, an outcome could be a 20% reduction in operational overhead by unifying IT processes. Grounding drivers in outcomes ensures the strategy focuses on delivering measurable value. Document these drivers and desired outcomes clearly, as they steer all subsequent decisions.

2. Create a clear vision statement for hybrid and multicloud

Craft a concise vision statement that describes the target state of your hybrid/multicloud environment and what success looks like. This vision statement provides direction and helps all stakeholders understand the end goal. The vision should articulate how the unified approach benefits the organization. For example:

- "Build an adaptive cloud platform that unifies all infrastructure and teams, enabling any app to run where it best meets business needs." (Emphasizes flexibility and unity.)
- "Deliver consistent customer experiences with 100% uptime through multicloud resilience." (Focuses on reliability and continuity.)
- "Increase deployment frequency by 50% by standardizing DevOps across cloud and on-premises." (Focuses on agility and process improvement.)
- "Reduce on-premises footprint by 50% in two years to cut costs, while extending cloud management to all remaining on-premises assets." (Emphasizes efficiency and cloud-first operations.)

3. Establish success metrics for hybrid and multicloud

Alongside the vision, define 2-5 key success metrics (KPIs) to measure progress. Each major driver from the previous step should map to at least one KPI. Make them specific and time-bound where possible. For example, if agility is a driver, a KPI might be provisioning time for infrastructure reduced from weeks to hours across all environments within 12 months. If cost optimization is a driver, a KPI might be to improve infrastructure utilization by 30% through cloud bursting and consolidation. Evaluate data egress and synchronization costs before adopting bursting. Include a security or compliance metric as well. For example, set a goal that 100% of onboarded and in-scope resources must be compliant with baseline security policies as measured via Azure Policy and Defender for Cloud.

By setting metrics, you create a shared definition of success. It aligns teams and provides a way to track the benefits of your hybrid/multicloud initiative over time.

4. Set principles for hybrid and multicloud

Establish guiding principles for deciding which cloud or environment to use for different workloads. Clear principles prevent random or preference-driven choices that could increase

complexity. They also balance the desire for portability against the benefits of cloud-specific services. Develop guidelines such as:

1. **Determine cloud-neutral versus cloud-specific usage.** Determine where you want to build cloud-agnostic solutions versus using cloud-native services. For each workload, decide if portability is critical. For example, for core systems of record you might prioritize neutrality, using technologies like Azure Kubernetes Service, containers, or databases that run anywhere. In contrast, for customer-facing or innovation projects, you might use cloud-specific PaaS services, like Azure Functions, to accelerate development.
2. **Use Azure as a unifying layer where possible.** Make Azure the central management and integration layer for all environments. Rather than maintaining parallel toolsets for each cloud, plan to manage other clouds through Azure. For example, if you run VMs in AWS, onboard them via Arc and enforce policies through Azure Policy to manage them like Azure resources. This unification ensures you don't have to use each cloud's native management tools. Azure becomes the consistent overlay across your entire IT estate.
3. **Justify multicloud complexity.** A multicloud approach can introduce complexity, such as multiple skill sets, varying architectures, and potentially higher costs like data egress charges between clouds. Set a principle that using another cloud must have a clear justification. Default to Azure for new deployments unless a unique capability or business requirement mandates otherwise. If a secondary cloud is used, integrate it through Azure and revisit its necessity periodically. This intentional stance avoids a sprawl of unmanaged, siloed deployments on different platforms. If a developer proposes using a different data analytics technology, they must have a strong reason and plan to integrate it. Otherwise, they should use Azure's analytics offerings like Microsoft Fabric.

These principles provide architects and engineers with a decision framework. For example, when choosing a database service, the guidelines might steer them to Azure's PaaS database for functionality rather than automatically picking a non-Azure service that doesn't align with strategy. The overall aim is to encourage using Azure as the backbone, minimize fragmentation, and only embrace multicloud elements when required.

5. Select your cloud mix for hybrid and multicloud

Decide which cloud platforms including on-premises should be part of your strategy, and establish Azure as the central management hub from the start:

1. **Choose your cloud portfolio based on requirements.** Evaluate business and technical needs to decide the mix of cloud platforms. Many enterprises use multiple public clouds to satisfy different needs or legacy investments. Document which platforms you should use and why. Also determine what role on-premises infrastructure plays going forward. For example, you might use Azure for most workloads and maintain on-premises systems for certain factory control systems using Azure Local. Be specific. Identify critical workloads that must remain on-premises and plan to modernize them by hosting on Azure Local and onboarding management via Azure Arc, plus any exceptional cases for other clouds. Ensure this planned mix ties back to your drivers, such as using a secondary cloud for resilience or specialized capabilities.
2. **Plan your cloud mix.** Planning your cloud mix means identifying current hybrid, edge, and multicloud environments, determining the mix that fits your future strategy, and deciding whether each platform operates independently or through unified operations. Once your intended cloud mix is defined, evaluate the configuration that best supports your environments and confirm the Azure landing zone option that fits your hosting approach. Customers distribute workloads differently: hybrid-first customers keep most workloads on-premises with selective use of edge, Azure, or other clouds; Azure-first customers host most in Azure with some on-premises or other environments; multicloud-first customers use another public cloud primarily while placing some workloads in Azure or at the edge. Your chosen mix should guide how each environment is configured and how it supports your overall cloud strategy.
3. **Make Azure your primary control plane for all environments.** Clearly state that Azure should serve as the main hub for managing resources across all clouds and on-premises. This decision is strategic because Azure offers strong hybrid management via Azure Arc and related services. By projecting AWS, Google Cloud, and on-premises resources into Azure for management, you centralize expertise and tooling. In practice, this means you should manage AWS resources from Azure using Arc rather than managing Azure resources from AWS's tools. Azure's cross-cloud services (Arc, Microsoft Defender for Cloud, Azure Monitor) make it well-suited for this hub role, providing integration and consistency.
4. **Design your unified operating model.** Envision how IT operations should function in this hybrid/multicloud setup. Define processes that work across all environments. For instance, your policy should be that all servers (Azure VMs, on-premises servers, AWS EC2) should be inventoried. They should also be configured via Azure Arc and governed by Azure Policy for Azure and Arc-enabled resources. For AWS or Google Cloud, use Defender for Cloud multicloud connectors to surface posture since compliance surfaces through Defender for

Cloud's multicloud connectors, not direct Azure Policy assignment. You should use CI/CD pipelines in GitHub Actions/Azure DevOps to deploy applications to any target environment using approved templates. Network operations should treat Azure as the hub linking other sites/clouds, and the security team should use Microsoft Sentinel to watch everything. By describing this target operating model, you set expectations that unified operations must replace silos, like separate ops teams for each cloud. It prepares the organization for changes in day-to-day work and clarifies how to achieve consistency.

- 5. **Establish unified teams to support cross-platform operations.** Alongside technology, plan the human aspect. Establish enabling teams to train and support platform and workload teams. Include members from traditional IT, cloud teams, and security. Train your on-premises IT staff in Azure and cloud skills so they can manage resources in any environment via Azure's tools. This approach shows that you expect collaboration and cross-training. In some cases, it might mean reorganizing. Maybe merging separate cloud teams or having a centralized platform that serves all business units. Making this part of the strategy ensures the organizational structure supports the operational model.

By deliberately selecting your cloud mix and choosing Azure as the anchor, you set a strong foundation for unified management. Everyone should understand what environments they should operate in and that Azure is how you tie them together. This understanding prevents uncontrolled proliferation of platforms and reinforces the earlier principles.

6. Map Azure hybrid and multicloud services to objectives

As you finalize the strategy, identify which Azure services and technologies help achieve your specific goals. This identification creates a bridge from high-level strategy to actionable implementation. Consider all key areas of your tech stack and map them to Azure solutions:

 Expand table

Technology area	Azure solution
Hybrid & multicloud management	Azure Arc – Project servers, supported Kubernetes clusters, Azure Local infrastructure, and selected data services into Azure to create a unified control plane. Centralize inventory, management, and policy enforcement across on-premises and other clouds where Arc support exists.
Identity & access management	Microsoft Entra ID – Use Entra ID as the unified identity platform across all environments via synchronization or federation. Provide single sign-on and centralized

Technology area	Azure solution
	credential management for Azure, on-premises AD, AWS, Google Cloud, and SaaS apps. Apply conditional access and role-based access control (RBAC) consistently.
Observability & monitoring	Azure Monitor – Consolidate logs, metrics, and traces from every environment into Azure Monitor. Use Log Analytics workspaces and Azure Monitor agents or Arc to ingest data from on-premises and other clouds.
Container orchestration	Azure Kubernetes Service (AKS) – Standardize on AKS for containerized workloads and manage clusters consistently using Arc-enabled Kubernetes. Use Azure Container Apps for serverless containers when an event-driven, scale-to-zero model fits.
Data & analytics	Microsoft Fabric – Create a unified analytics layer that connects on-premises SQL, Azure data lakes, and third-party cloud sources into a single data estate.
IoT & edge computing	Azure IoT Hub and Azure IoT Edge – Manage devices and run edge processing with Azure IoT services. Integrate IoT deployments with Azure Arc to enforce governance and secure edge compute while onboarding devices to the unified management plane.
On-premises infrastructure	Azure Local – Run VMs and selected Azure services on new on-premises hardware or private cloud using Azure Stack HCI. Integrate those systems with Azure via Arc for consistent management and policy control.
Security & governance	Microsoft Defender for Cloud – Use Defender for Cloud to manage cloud security posture and protect workloads across Azure, AWS, and GCP. Combine Azure Policy to enforce policies on Azure and Arc-enabled resources, and use Microsoft Sentinel for SIEM and SOAR across logs from all environments.

Documenting this mapping ensures your strategy includes a concrete Azure-centric technology game plan. It also helps identify skill gaps and tooling needs early. For example, if you plan to use Microsoft Fabric for analytics, you know you need data integration skills and Power BI expertise. If Azure Arc is central, you plan training for your ops teams on Arc. This step translates your strategic intentions into actionable Azure services that make it happen.

Strategy outcome

At the end of this phase, you should have a hybrid/multicloud strategy that captures all the above elements. It should summarize your decisions so far:

- **Business drivers & scope:** Unify IT operations, uptime requirements for multicloud, edge requirements.
- **Vision statement:** Azure is the primary platform and control plane, integrating other clouds and on-premises systems to provide a unified, agile digital infrastructure.
- **Success metrics:** Specific targets for availability, deployment speed, cost savings, and compliance.

- **Guiding principles:** Stances on avoiding lock-in, defaulting to Azure, using cloud-neutral designs where needed.
- **Cloud mix:** Which environments are used (Azure, on-premises via Azure Local, other clouds) and why.
- **Key technologies:** The Azure and any non-Azure services decided upon to execute the strategy.

To illustrate, here's an example excerpt of how a strategy might be summarized:

- **Example strategy summary:** The organization pursues an adaptive hybrid/multicloud approach to unify IT operations while using the best cloud capabilities for each need.
- **Drivers:** Avoid downtime (target < 1 hour/year). Meet EU data residency laws. Break down ops silos to improve efficiency (target 20% OPEX reduction).
- **Vision:** Azure is our primary cloud and control plane integrating all other environments. We must improve global coverage using Azure plus one secondary cloud and Azure Local for on-premises needs.
- **Cloud principles:** Embrace Azure-native services for differentiation and speed. Default new deployments to Azure unless a compelling requirement dictates otherwise.
- **Cloud mix:** Currently ~50% on-premises to be modernized with Azure Local, 40% Azure, 10% AWS for a specific use case. Long-term aim: 70% Azure, with remaining on-premises running via Azure Local and only niche use of other clouds.
- **Key Azure technologies:** Azure Arc to unify resource management. Azure Monitor and Defender for end-to-end visibility and security. Azure Local to extend cloud to on-premises. AKS for containers. Microsoft Fabric to unify data analytics. Azure IoT for edge devices. Entra ID for unified identity. Standardized Azure Pipelines for all deployments. We must measure success by our ability to meet uptime and deployment frequency targets while maintaining strict security compliance across all platforms.

Having this strategic brief approved by stakeholders ensures everyone is aligned before moving on. Now you can proceed to detailed planning with a solid Azure-focused game plan in hand.

Next steps

Ready your environment

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